<u>**Questions to analyse**</u>

1. Is there a statistically significant linear correlation between high temperatures and humidity levels across different climate zones?
2. How strongly does wind speed correlate with changes in atmospheric pressure?
3. Which variables (e.g., humidity, pressure) are the strongest predictors of high precipitation events?
4. How have average monthly temperatures shifted globally over the recorded period?
5. Are there recurring seasonal patterns in precipitation that vary significantly between the Northern and Southern Hemispheres?
6. Which regions experience the highest daily temperature swings (diurnal range)?
7. How do average wind speeds in coastal regions compare to inland or mountainous climate zones?
8. Which specific countries or continents are currently showing the highest rate of temperature increase?
9. Which geographic regions are identified as "high-precipitation zones" during specific months?
10. What are the top 1% of extreme wind speed events recorded, and where did they occur?
11. Can we identify specific dates where temperatures deviated by more than two standard deviations from the historical mean for a region?
12. Are there regions experiencing "flash" precipitation events that fall outside of expected seasonal trends?
13. In which specific climate zones does the correlation between temperature and humidity break down (become inverse), and what does this suggest about local "dry heat" vs. "humid heat" trends?
14. Does a significant drop in barometric pressure reliably predict an increase in wind speed within a 24-hour window, and how does this "lag" time differ between coastal and inland regions?
15. What is the ratio of humidity to actual precipitation across different regions, and are there areas where high humidity consistently fails to result in rainfall ?
16. How often do "compound extremes"—such as simultaneous high heat and high wind speeds—occur in the dataset, and has the frequency of these combined events increased over the recorded period?
17. If we define "extreme" as being 3 standard deviations 3σ from the regional mean, which continents are seeing the fastest growth in the *number* of these outliers for temperature and precipitation?
18. After a peak "extreme weather event" (like a storm or heatwave), what is the average "recovery time" for a region to return to its 30-day rolling mean temperature?

19. Is the transition period between seasons (e.g., Winter to Spring) becoming shorter or more volatile? You can measure this by the "rate of change" in weekly average temperatures.
20. Is the gap between daily maximum and minimum temperatures (Diurnal Temperature Range) narrowing globally, and does this correlate with increased cloud cover or humidity in those regions?
21. Are weather patterns (like 5+ day streaks of above-average heat) becoming more "persistent" over time compared to historical fluctuations in the dataset?
22. Are the weather profiles (temperature/humidity signatures) of certain regions beginning to more closely resemble a different climate zone than they were historically categorized in?
23. Which geographic regions show the highest "standard deviation" in their weather variables, making them the most unpredictable for local decision-making and agriculture?